Configure Your Dataverse Repository to Register DataCite DOIs

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The Dataverse® Project

IQSS
The Institute for Quantitative Social Science

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Information Technology

Agenda

1. The Dataverse Project
2. Dataverse and PIDs
3. Metadata for Discoverability and Reuse
4. Global Dataverse Community Consortium (GDCC), Dataverse, and DataCite
5. How Dataverse + DataCite work together

The Dataverse® Project

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The Dataverse Project

The Dataverse Project



Platform & Purpose

- An open-source platform to publish, cite, and archive research data
- Built to support multiple types of data, users, and workflows



Origins & Support

- Developed at Harvard's Institute for Quantitative Social Science (IQSS) since 2006
- Development funded by IQSS and with grants, in collaboration with institutions around the world



Core Team & Community

- Core team
  - @ IQSS - developers, UX/UI, metadata specialists, curation team, leadership team
  - key contributors from the community with full privileges as IQSS team

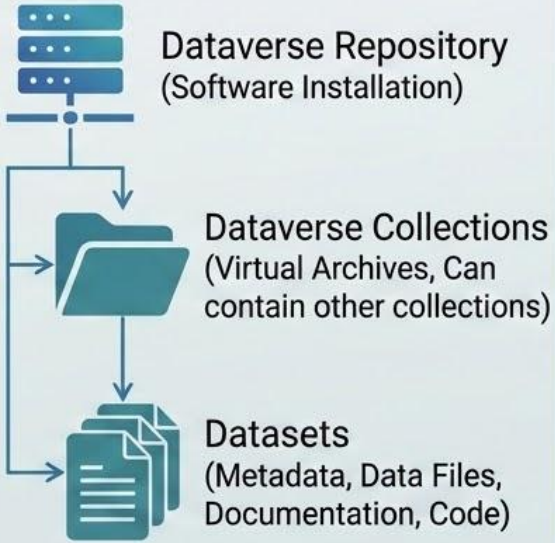
The Dataverse Project

Overview & Key Concepts

What it is & Benefits

- Share & Preserve Data
- Cite, Explore & Analyze
- Replicate Work Easily
- Academic Credit & Web Visibility

Structure & Organization



The Central Insight & Solution

Problem: Credit (Distribution) vs. Preservation (Archives)

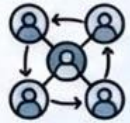


Solution: Dataverse provides both. Institutional Backing, Long-term Preservation + Full Credit & Web Visibility.

See King, G. (2007). An Introduction to the Dataverse Network... *Sociological Methods & Research, 36*(2)...

A major international collaborative project. We encourage you to join us.

Dataverse - The Strategic Goals



Grow Community

Expand the user & contributor base



Empower Open Source

Enable new applications, tools & services



Handle Sensitive & Big Data

Develop capabilities for secure & large-scale data



Expand Data Features

Support existing & new disciplines



Expand Archival Features

Enhance long-term preservation



Increase Interoperability

Implement standards for seamless exchange



Increase Contributions

Boost open source development



Improve User Experience

Refine usability & accessibility

The Dataverse Project

Dataverse Repositories - A World View

View more Metrics



Dataverse and PIDs

Dataverse and PIDs

Dataverse supports PIDs at the dataset and file level.



Function:
Unique, actionable pointers for assets (datasets, pubs) or entities (researchers, orgs).

Persistent Linkage:
Managed via registries, ensuring validity even if resources move.

Types of PIDs:
- DOI (Digital Object Identifier)
- ORCID (Open Researcher and Contributor ID)
- ROR (Research Organization Registry)

Relationship to Metadata: PIDs are accompanied by metadata to provide context (author, title, date), facilitating machine-actionable principles.

Core PIDs to Power Findability

1. Digital Object Identifiers (DOIs)
2. Open Researcher and Contributor (ORCID) IDs
3. Research Organization Registry (ROR) IDs



Dataverse and DataCite - the connection

March 30, 2016

Key Announcement

Dataverse 4.3 (mid-March) enables direct DOI minting via DataCite Metadata Store API.

Enhanced Discovery

Supports more metadata: description, author affiliation, ORCID identifier.

DataCite's Perspective

Patricia Cruse, Executive Director: "Sharing data is good. Dataverse is... a critical component... connecting communities... accelerate data sharing."

Collaboration & Acknowledgment

Help from Peking University (open source).

Visit from Martin Fenner, DataCite Technical Director, to design the feature.

Newest Update

Phase 2 completed: Multiple institutions with own DataCite DOI prefixes (GitHub Issue #2943).

Metadata for Discoverability and Reuse

What is “good” metadata?



Metadata that helps others find datasets

- Clear Title & Abstract
- Keywords & Subjects
- Author & Affiliation



Metadata that helps others re-use datasets

- License & Terms of Use
- Methodology & Data Dictionary
- Linked Publications & Code

What is “good” metadata?

Metadata that helps others find datasets

Metadata fields that discovery systems use when returning search results



DataCite

OpenAIRE

Metadata that leverages how people look for data

DDI®

schema.org

Metadata that helps others find datasets

- Metadata fields that discovery systems use when returning search results
- Metadata that leverages how people look for data



Metadata that helps others re-use datasets

Terms of use and access metadata, e.g. CC0

Links to other resources, e.g. “Related Publications” metadata

Readme files



Readme files

Global Dataverse Community Consortium (GDCC) and DataCite

GDCC, Dataverse, and DataCite

The Global Dataverse Community Consortium (GDCC) acts as a DataCite consortium, allowing member organizations using Dataverse software to manage DOIs.



Membership Benefits:

- Reduced Dataverse community event rates
- Support for repository installation
- Collective DOI management

DOI Management: Through the GDCC consortium, organizations can register DOIs efficiently.

GDCC, Dataverse, and Datacite



DOI Registration

Dataverse integrates directly with DataCite to register DOIs, ensuring datasets are citable.



Automatic Metadata Updates

When datasets are published, Dataverse pushes metadata to DataCite to improve visibility.



Configuration

Dataverse installation administrators configure DOI providers (typically DataCite) using API credentials.



OAI-PMH Support

Dataverse supports harvesting metadata directly from DataCite using the OAI-PMH protocol.

GDCC, Dataverse and DataCite

Members

We encourage organizations using the Dataverse software to become a member of GDCC.

Your GDCC membership helps us

- coordinate community efforts through interest groups and working groups, pull requests, resources provided through the GDCC GitHub Repositories;
- support the community on technical and organizational issues through the GDCC Dataverse Repository Installation and Consulting Services, through GDCC support in discussion groups, on Slack, and other channels;
- provide DOIs through GDCC consortium membership in DataCite;
- organize community events.

provide DOIs through GDCC consortium membership in DataCite;

organize community events.

As a GDCC member organization, you will profit from the above benefits. In addition, your employees will profit from reduced rates at Dataverse community events.

We welcome applications from all organizations.

Connecting Dataverse + DataCite

Setting up DataCite DOIs in Dataverse

Dataverse allows for multiple PID providers, so the first step is to set some global settings that tell Dataverse about any (and all) providers

- Dataverse.pid.providers
  - A comma-separated list of the ids of the PID providers to use. IDs should be simple unique text strings, e.g. datacite1, perma1, etc. IDs are used to scope the provider-specific settings but are not directly visible to users.
- Dataverse.pid.default-provider
  - The ID of the default PID provider to use.

Setting up DataCite DOIs in Dataverse (continued)

Per-Provider Settings:

Each Provider listed by id in the dataverse.pid.providers setting must be configured with the following common settings and any settings that are specific to the provider type.

- dataverse.pid.*.type
  - The Provider type, currently one of `datacite`, `ezid`, `FAKE`, `hdl`, or `perma`. The type defines which protocol a service supports (DOI, Handle, or PermaLink) and, for DOI Providers, which DOI service is used.
- dataverse.pid.*.label
  - A human-readable label for the provider
- dataverse.pid.*.authority
- dataverse.pid.*.shoulder
  - In general, PIDs are of the form `<protocol>:<authority>/<shoulder>*` where `*` is the portion unique to an individual PID. PID Providers must define the authority and shoulder (with the protocol defined by the `dataverse.pid.*.type` setting) that defines the set of existing PIDs they can manage and the prefix they can use when minting new PIDs. (Often an account with a PID service provider will be limited to using a single authority/shoulder. If your PID service provider account allows more than one combination that you wish to use in Dataverse, configure multiple PID Provider, one for each combination.)
-

Setting up DataCite DOIs in Dataverse (continued)

DataCite specific Settings:

PID Providers of type datacite require four additional parameters that define how the provider connects to DataCite. DataCite has two APIs that are used in Dataverse:

- dataverse.pid.*.datacite.mds-api-url
  - Configure the base URL of the DataCite MDS API, used to mint and manage DOIs.
- dataverse.pid.*.datacite.rest-api-url
  - Configure the base URL endpoint of the DataCite REST API, used for PIDs API.
- dataverse.pid.*.datacite.username
  - DataCite uses HTTP Basic authentication for Fabrica and their APIs. You need to provide the same credentials to Dataverse software to mint and manage DOIs for you.
- dataverse.pid.*.datacite.password
  - Once you have a password from your provider, you should create a password alias. This avoids storing it in clear text, although you could use a JVM option to reference a different place.

Setting up DataCite DOIs in Dataverse (continued)

Optional Settings:

- dataverse.pid.*.identifier-generation-style
    - By default, the Dataverse Software generates a random 6 character string
    - Set this to `storedProcGenerated` to generate instead a custom unique identifier through a database stored procedure or function (Note: In addition to this setting, a stored procedure or function must be created in the database)
- dataverse.pid.*.datafile-pid-format
    - This setting controls the way that the “identifier” component of a file’s persistent identifier (PID) relates to the PID of its “parent” dataset.
    - By default the identifier for a file is dependent on its parent dataset
        - For example (for dataset TJCLKP): TJCLKP/MLGWJO
    - Alternatively, the identifier for File PIDs can be configured to be independent of Dataset PIDs using the setting INDEPENDENT

Registering PIDs for files in select collections

- It is possible to configure registering DataCite DOIs for files in individual collections.
  - registration of DOIs for files can be enabled in a specific collection when it is disabled instance-wide
  - Or it can be disabled in specific collections where it is enabled by default
- Available via the Admin API

:FilePIDsEnabled

Toggles publishing of file-level PIDs for the entire installation. By default this setting is absent and Dataverse Software assumes it to be false. If enabled, the registration will be performed asynchronously (in the background) during publishing of a dataset.

It is possible to override the installation-wide setting for specific collections, see :AllowEnablingFilePIDsPerCollection. For example, registration of PIDs for files can be enabled in a specific collection when it is disabled instance-wide. Or it can be disabled in specific collections where it is enabled by default. See Change Collection Attributes for details.

To enable file-level PIDs for the entire installation:

```
``curl -X PUT -d 'true' http://localhost:8080/api/admin/settings/:FilePIDsEnabled``
```

If you don't want to register file-based PIDs for your entire installation:

```
``curl -X PUT -d 'false' http://localhost:8080/api/admin/settings/:FilePIDsEnabled``
```

:AllowEnablingFilePIDsPerCollection

Toggles whether superusers can change the File PIDs policy per collection. By default this setting is absent and Dataverse Software assumes it to be false.

For example, if this setting is true, registration of PIDs for files can be enabled in a specific collection when it is disabled instance-wide. Or it can be disabled in specific collections where it is enabled by default. See Change Collection Attributes for details.

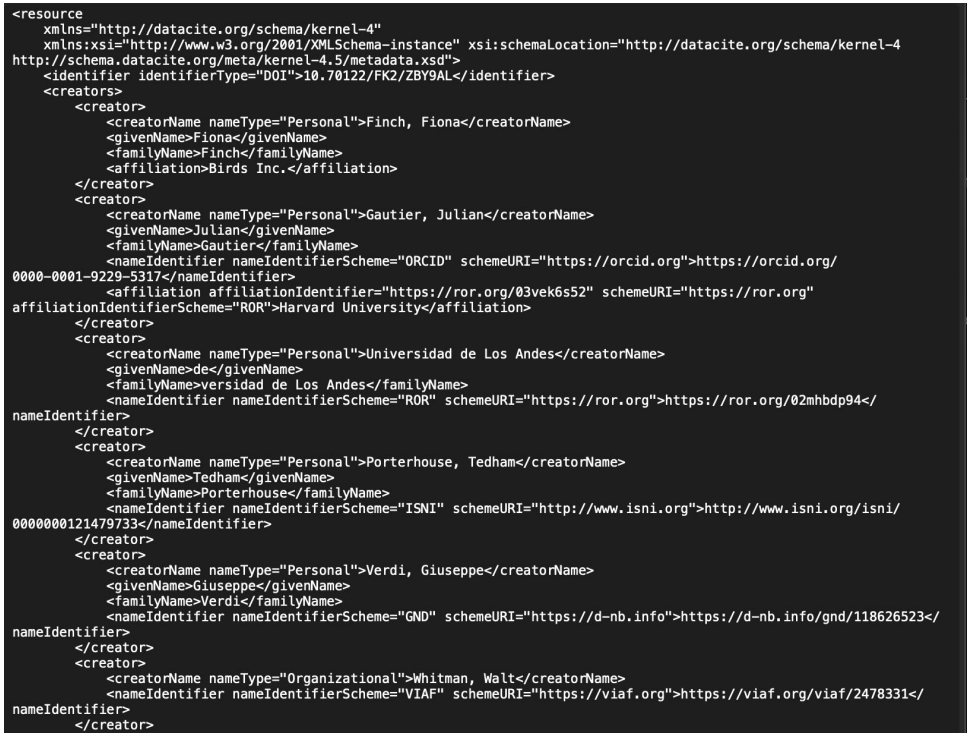
To enable setting file-level PIDs per collection:

```
``curl -X PUT -d 'true' http://localhost:8080/api/admin/settings/:AllowEnablingFilePIDsPerCollection``
```

When :AllowEnablingFilePIDsPerCollection is true, setting File PIDs to be enabled/disabled for a given collection can be done via the Native API - see Change Collection Attributes in the Native API Guide.

DataCite Export

- When publishing a dataset via Dataverse, Dataverse exports include **DataCite Metadata Schema for the Publication and Citation of Research Data and Other Research Outputs. Version 4.5**
  - This includes Citation Metadata, Geospatial Metadata, License information, etc.
  - Dataverse Metadata Crosswalk



```
<resource
    xmlns="http://datacite.org/schema/kernel-4"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datacite.org/schema/kernel-4
http://schema.datacite.org/meta/kernel-4.5/metadata.xsd">
    <identifier identifierType="DOI">10.70122/FK2/ZBY9AL</identifier>
    <creators>
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        </creator>
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            <nameIdentifier nameIdentifierScheme="VIAF" schemeURI="https://viaf.org">https://viaf.org/viaf/2478331</
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        </creator>
```

OAI-PMH Harvesting from DataCite

- DataCite's OAI gateway (https://oai.datacite.org/oai) serves records for every DOI they have registered, so we can **harvest metadata from any participating institution** even if they do not maintain an OAI server of their own
- Their OAI implementation offers a concept of a "dynamic set", making it possible to use **any query supported by the DataCite search API** as though it were a "set".
    - This makes harvesting from them extra flexible, allowing users to harvest virtually any arbitrary subset of metadata records, potentially spanning multiple institutions and registration authorities.
- Learn more: https://guides.dataverse.org/en/latest/admin/harvestclients.html#harvesting-from-datacite



The Collaboration



IQSS
(Institute for Quantitative Quantitative Social Science) - Leads open source software development.



Harvard Library & HUIT
- Harvard Dataverse Repository hosting, backups, and making installation openly available.

The Institute for Quantitative Social Science (IQSS) collaborates with the Harvard University Library and Harvard University Information Technology organization to make the installation of the Harvard Dataverse Repository openly available to researchers and data collectors worldwide from all disciplines, to deposit data. IQSS leads the development of the open source Dataverse Project software and, with the Open Data Assistance Program at Harvard (a collaboration with Harvard Library, the Office for Scholarly Communication and IQSS), provides user support. The Library Technology Services at HUIT provides hosting and backups support of the Harvard Dataverse Repository. The Dataverse Project also collaborates with the Global Dataverse Community Consortium to help support Community needs.

Open Data Assistance Program (Harvard Library, OSC, IQSS)
- Provides user support.





The Dataverse Project



Global Dataverse Community Consortium
- Collaborates to support Community needs.

Thank You

References

The Dataverse Project

The Dataverse Guide

Introduction to Metadata **in Dataverse Repositories: For Researchers and Support Staff**

Contact us: Support@dataverse.org